

01) — CEH v13

Module 01

1. **Information Security Basics** — CIA Triad, AAA, Security Controls
2. **Ethical Hacking Concepts** — Types of Hackers, Phases of Hacking
3. **Legal Framework** — Laws, Compliance, Cyber Crimes
4. **Lab Setup** — Kali Linux, VMs, Target machines
5. **Linux Terminal Basics** — Essential commands for pentesting

CIA Triad — Information Security ke 3 Pillars

Component	Full Form	Matlab
C	Confidentiality	Data sirf authorized logon ko mile — Encryption, Access Controls
I	Integrity	Data change/alter nahi hona chahiye — Hashing, Checksums
A	Availability	Data aur services hamesha available hon — Redundancy, Backups

♦ **Example:** Army ka encrypted communication channel — sirf authorized personnel access kare (Confidentiality), message beech mein badla na jaaye (Integrity), network hamesha up rahe (Availability).

DAD Model (CIA ka opposite — Attack Perspective)

Letter	Attack	Example
D	Disclosure (Confidentiality break)	Data leak, hacking
A	Alteration (Integrity break)	Man-in-the-middle, data tampering
D	Destruction (Availability break)	DoS/DDoS attack

AAA Framework

Term	Full Form	Meaning
Authentication	Who are you?	Username/password, biometric, smart card

Term	Full Form	Meaning
Authorization	What can you do?	Access levels, permissions
Accounting	What did you do?	Logs, audit trails

📌 Security Controls

Type	Examples
Physical	CCTV, locks, biometric gates, security guards
Technical	Firewalls, IDS/IPS, encryption, antivirus, SIEM
Administrative	Policies, SOPs, NDA, security awareness training

📌 Types of Hackers (CEH Exam mein aata hai)

Category	Description
White Hat	Ethical hacker — authorized, legal, finds vulns to fix
Black Hat	Criminal hacker — unauthorized, illegal, personal gain
Gray Hat	Unauthorized but no malicious intent — finds vuln, discloses it
Script Kiddie	Pre-made tools use karta hai, no technical knowledge
Hactivist	Political/social cause ke liye hack
State-Sponsored	Government ke liye hack — espionage, warfare
Suicide Hacker	Getting caught se koi fark nahi padta
Cyber Terrorist	Fear spread karna, critical infra attack

📌 Phases of Ethical Hacking (CEH Methodology — 5 Phases)

Phase	Kya Hota Hai	Tools/Techniques
1. Reconnaissance	Target ke baare mein info collect	Passive: Google, WHOIS, Social media. Active: Nmap ping scan

Phase	Kya Hota Hai	Tools/Techniques
2. Scanning & Enumeration	Live hosts, open ports, services identify	Nmap, Netcat, enum4linux
3. Gaining Access	Vulnerability exploit karke entry lena	Metasploit, SQL injection, password cracking
4. Maintaining Access	Backdoors, RATs, persistent access	Meterpreter, Trojans, rootkits
5. Covering Tracks	Logs delete, evidence hide	clearev, wevtutil, log tampering

Types of Penetration Testing

Type	Description
Black Box	Zero knowledge of target — real attacker simulation
White Box	Full knowledge — source code, network diagrams available
Gray Box	Partial knowledge — limited credentials / info

Cyber Laws & Compliance

Law/Standard	Region	Purpose
IT Act 2000	India	Cyber crimes, digital signatures
GDPR	Europe	Data privacy protection
HIPAA	USA	Healthcare data protection
PCI DSS	Global	Payment card data security
ISO 27001	Global	Information security management
DMCA	USA	Copyright/Digital content

(Lab Setup + Linux Basics)

Step 1: Kali Linux Install Karna

bash

ISO download: <https://www.kali.org/get-kali/>

VirtualBox/VMware mein new VM banao

4GB RAM, 2 CPU cores, 60GB disk minimum

Boot from ISO → Graphical Install

Username: kali / Password: kali

Step 2: Target Machines Install

Metasploitable 2 (Linux Target):

bash

Download: <https://sourceforge.net/projects/metasploitable/>

VirtualBox mein import karo (File → Import Appliance)

Login: msfadmin / msfadmin

Windows 10 VM (Windows Target):

bash

Windows 10 ISO download karo

Normal installation

User: admin / Password: Password123

Remote Desktop enable karo

Windows Firewall disable karo (lab ke liye)

Step 3: Network Configuration

VirtualBox Network Settings:

Kali Linux → NAT Network ya Host-Only Adapter

Metasploitable → Same network as Kali

Windows VM → Same network as Kali

Connectivity test (sab machines se):

bash

Kali se:

ping 192.168.1.20 *# Metasploitable*

ping 192.168.1.30 *# Windows*

Metasploitable se:

ifconfig *# Apna IP dekho*

ping 192.168.1.10 *# Kali ko ping*

Windows se:

ipconfig *# Apna IP dekho*

ping 192.168.1.10 *# Kali ko ping*

Linux Terminal Basics — Pentesting

File System Commands

bash

pwd *# Current directory dikhata hai (Print Working Directory)*

ls -la *# Directory list — detailed view + hidden files*

cd /etc *# Directory change*

mkdir projects *# Folder banao*

touch notes.txt *# Empty file banao*

cp notes.txt backup.txt *# Copy file*

mv notes.txt /tmp/ *# Move/rename file*

rm notes.txt *# File delete*

rm -rf /tmp/test/ *# Folder + content delete (⚠ careful!)*

cat /etc/passwd *# File content dekho*

less /etc/passwd *# Page-by-page view (q se exit)*

head -20 /etc/passwd *# Pehli 20 lines*

tail -20 /var/log/syslog *# Aakhri 20 lines*

nano file.txt *# Text editor (Ctrl+X to exit, Y to save)*

Permission Commands

bash

chmod 777 file *# Full permission (r=4, w=2, x=1) — owner/group/others*

chmod 755 file *# Owner:rwx, Group:r-x, Others:r-x*

chown user:group file *# Owner/group change*

umask *# Default permission mask dekho*

Network Commands

bash

ifconfig *# Network interfaces dekho (IP, MAC)*

ip addr *# Modern replacement of ifconfig*

ip a *# Shortcut (same as above)*

iwconfig *# Wireless interfaces info*

ping 192.168.1.1 *# Host alive hai ya nahi check*

ping -c 4 192.168.1.1 *# Sirf 4 pings bhejo*

traceroute 8.8.8.8 *# Route trace karo*

traceroute -n 8.8.8.8 *# IP address show kare (no DNS)*

netstat -tulpn *# Open ports dekho*

netstat -an | grep LISTEN *# Listening services*

ss -tulpn *# Modern netstat (faster)*

arp -a *# ARP table dekho (neighbors)*

route -n *# Routing table*

User & System Info

bash

whoami *# Current user*

id *# User ID, Group ID*

uname -a *# System info (kernel, architecture)*

hostname *# Machine name*

who *# Currently logged in users*

w *# Who + what they're doing*

last *# Last login logins*

uptime *# System kitne time se chal raha*

free -h *# RAM usage*

df -h *# Disk usage*

ps aux *# Running processes*

top *# Real-time process monitor (q to quit)*

Finding Things

bash

find / -name "*.txt" 2>/dev/null *# Files search karo*

find / -perm -4000 2>/dev/null *# SUID files (privesc ke liye)*

grep -r "password" /etc/ *# Files ke andar text search*

which nmap *# Tool kahan installed hai*
locate hydra *# Database se file find (updatedb pehle)*

Package Management

bash
apt update *# Package list update*
apt upgrade *# All packages upgrade*
apt install nmap *# Tool install*
apt remove nmap *# Tool remove*
apt search snmp *# Package search*
dpkg -l | grep nmap *# Installed hai ya nahi check*

File Download

bash
wget https://example.com/file.zip *# Web se file download*
curl -O https://example.com/file.zip *# Alternative download tool*
curl -I https://example.com *# Sirf headers dekho*

Essential Pentesting Tools — Installation Check

bash
In sab tools ko check karo ki installed hain ya nahi
which nmap
which metasploit *# Ya: /usr/bin/msfconsole*
which hydra
which john
which sqlmap

which wireshark

which burpsuite *# Ya: /usr/bin/burpsuite*

which aircrack-ng

which gobuster

which dirb

which whatweb

which nikto

which enum4linux

which snmpwalk

which netcat *# ya: which nc*

which nslookup

which dig

which dnsrecon

which theharvester

which recon-ng

which ettercap

which setoolkit

Missing tool hai to install karo:

bash

apt install -y <tool-name>



CEH Lab Setup Checklist (Day 1 Complete Karne ke Baad)

☒	Item	Status
☐	Kali Linux installed & booting	

☒	Item	Status
☐	Metasploitable 2 installed	
☐	Windows VM installed	
☐	All 3 VMs same network pe hain	
☐	Kali → Metasploitable ping ho raha	
☐	Kali → Windows VM ping ho raha	
☐	All essential tools installed (nmap, wireshark, metasploit, etc.)	
☐	Basic Linux commands practice ho gayi	

Practical Assignments — Day 1

Assignment 1: Network Connectivity Test

bash

Kali se dono targets ko ping karo

ping -c 4 192.168.1.20

ping -c 4 192.168.1.30

Output save karo

ping -c 4 192.168.1.20 > ping_results.txt

cat ping_results.txt

Assignment 2: Network Interface Info Collect

bash

Apne machine ka network info collect karo

ifconfig

ip addr show

arp -a

```
route -n
```

```
cat /etc/resolv.conf
```

```
# Output file mein save karo
```

```
ifconfig > my_network_info.txt
```

```
cat my_network_info.txt
```

Assignment 3: System Info Collect

```
bash
```

```
# Target machine ke baare mein info collect karo
```

```
uname -a
```

```
cat /etc/os-release # OS version
```

```
cat /proc/cpuinfo # CPU info
```

```
free -h # RAM
```

```
df -h # Disk
```

```
whoami
```

```
id
```

```
hostname
```

Assignment 4: File System Practice

```
bash
```

```
# Ek directory banao
```

```
mkdir ~/ceh_lab
```

```
cd ~/ceh_lab
```

```
# Ek file banao jisme target info ho
```

```
echo "Target IP: 192.168.1.20" > target_notes.txt
echo "Target OS: Metasploitable 2" >> target_notes.txt
cat target_notes.txt
```

File copy karo

```
cp target_notes.txt target_backup.txt
```

Directory structure banao

```
mkdir -p ~/ceh_lab/{recon,scanning,exploitation,post-exploitation,reporting}
```

```
ls -la ~/ceh_lab/
```

Assignment 5: Tool Availability Check Script

```
bash
```

```
#!/bin/bash
```

File banao: check_tools.sh

chmod +x check_tools.sh

./check_tools.sh

```
TOOLS="nmap msfconsole hydra john sqlmap wireshark burpsuite aircrack-ng gobuster
dirb whatweb nikto enum4linux snmpwalk netcat nslookup dig dnsrecon theharvester
recon-ng ettercap setoolkit"
```

```
echo "Checking essential CEH tools..."
```

```
echo "=====
```

```
for tool in $TOOLS; do
```

```
    if which $tool &>/dev/null; then
```

```
echo "✅ $tool - FOUND at $(which $tool)"  
else  
    echo "❌ $tool - NOT FOUND"  
fi  
done  
  
echo "=====  
echo "Check complete!"
```

CEH v13 Exam — Module 01 se Related Questions (Practice)

1. **CIA Triad ka kya matlab hai?**
2. **White Hat vs Black Hat hacker mein kya difference hai?**
3. **Ethical Hacking ke 5 phases kya hain?**
4. **Types of penetration testing kya hain?**
5. **AAA framework kya hai?**
6. **GDPR kya protect karta hai?**
7. **Hacktivist kaun hota hai?**
8. **Physical security controls ke 3 examples do**
9. **CIA Triad ka opposite (DAD model) kya hai?**
10. **Gray Hat hacker kaun hota hai?**